NETW1005 Project

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Lab wednesday
52-5930

(a) lsblk -fs

```
[student@server3 ~]$ lsblk -fs
NAME          FSTYPE     LABEL          UUID                                   MOUNTPOINT
sda1          xfs                       d695fa3d-7f9a-4e7a-98ae-031e672b4d64   /boot
└─sda
sdb1          xfs                       d632c32b-38c0-4756-9d8a-a8348e0ff255   /basic_storage_1
└─sdb
sdb2          swap                      8d775690-77fa-48bb-ad8c-64981bbbd349
└─sdb
sdb3          swap                      2ba6b636-98b7-4a9e-bc19-af9279e3f2ff   [SWAP]
└─sdb
sr0           iso9660    VBox_GAs_7.1.6 2025-01-21-13-18-27-98                 /run/media/student/VBox_GAs_7.1.6
rhel-root     xfs                       28a81415-ffc1-4697-8716-4aa7005f809a   /
└─sda2        LVM2_membe                1hXCiK-XmWX-9M04-qOun-An9f-Lu2v-uDi6u4
  └─sda
rhel-swap     swap                      4b2a8329-e04f-4e51-9fe2-fc9b087aa8cd   [SWAP]
└─sda2        LVM2_membe                1hXCiK-XmWX-9M04-qOun-An9f-Lu2v-uDi6u4
  └─sda
vg_lvm_1-lv_1 xfs                       13b1a9ed-fcc9-435e-af5e-a74545101a2d   /lvm_1
├─sdb4        LVM2_membe                g0d3vd-TBok-Hw6n-vXg8-9mZo-UThk-XTkj7x
│ └─sdb
├─sdb5        LVM2_membe                jHKf3E-nz8j-Lkpl-KFZ4-Xg6e-xUUR-sPzQ0c
│ └─sdb
└─sdb6        LVM2_membe                j3E0HS-P4HB-GvmE-2u8L-5V7S-0dtT-XCflsh
  └─sdb
vg_data-lv_ingestion
│             ext4                      91daa08e-6cb9-499e-9b93-7c4b36eec350   /root/ingestion
└─sdc         LVM2_membe                d2Hkr0-Kntz-53M1-B7dY-Za3I-Y2ya-GisBlc
vg_data-lv_analyses
│             ext4                      d7342589-52c2-49bf-9cd7-152339fabdaa   /root/analyses
└─sdc         LVM2_membe                d2Hkr0-Kntz-53M1-B7dY-Za3I-Y2ya-GisBlc
[student@server3 ~]$
```

```
[student@server3 ~]$ lsblk
NAME                  MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda                     8:0    0   30G  0 disk
├─sda1                  8:1    0    1G  0 part /boot
└─sda2                  8:2    0   29G  0 part
  ├─rhel-root         253:0    0   26G  0 lvm  /
  └─rhel-swap         253:1    0    3G  0 lvm  [SWAP]
sdb                     8:16   0    2G  0 disk
├─sdb1                  8:17   0  500M  0 part /basic_storage_1
├─sdb2                  8:18   0  100M  0 part
├─sdb3                  8:19   0  100M  0 part [SWAP]
├─sdb4                  8:20   0  100M  0 part
│ └─vg_lvm_1-lv_1     253:2    0  588M  0 lvm  /lvm_1
├─sdb5                  8:21   0  200M  0 part
│ └─vg_lvm_1-lv_1     253:2    0  588M  0 lvm  /lvm_1
└─sdb6                  8:22   0  300M  0 part
  └─vg_lvm_1-lv_1     253:2    0  588M  0 lvm  /lvm_1
sdc                     8:32   0    8G  0 disk
├─vg_data-lv_ingestion 253:3    0    2G  0 lvm  /root/ingestion
└─vg_data-lv_analyses  253:4    0  1.5G  0 lvm  /root/analyses
sr0                    11:0    1 57.4M  0 rom  /run/media/student/VBox_GAs_7.1.6
[student@server3 ~]$
```

(b) cat /etc/fstab

```
[student@server3 ~]$ cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Wed Aug 19 10:06:24 2020
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/rhel-root   /                       xfs     defaults        0 0
UUID=d695fa3d-7f9a-4e7a-98ae-031e672b4d64 /boot                   xfs     defaults        0 0
/dev/mapper/rhel-swap   swap                    swap    defaults        0 0
UUID=d632c32b-38c0-4756-9d8a-a8348e0ff255 /basic_storage_1 xfs defaults 0 0
UUID=2ba6b636-98b7-4a9e-bc19-af9279e3f2ff swap swap pri=20 0 0
UUID=13b1a9ed-fcc9-435e-af5e-a74545101a2d /lvm_1 xfs defaults 0 0
UUID=d7342589-52c2-49bf-9cd7-152339fabdaa /root/analyses ext4 defaults 0 0
UUID=91daa08e-6cb9-499e-9b93-7c4b36eec350 /root/ingestion ext4 defaults 0 0
[student@server3 ~]$
```

(c) findmnt -verify

```
[root@server3 ~]# findmnt --verify
/
   [W] recommended root FS passno is 1 (current is 0)
swap
   [W] target specified more than once

0 parse errors, 0 errors, 2 warnings
[root@server3 ~]#
```

(d) pvdisplay , vgdisplay , lvdisplay

```
[root@server3 ~]# pvdisplay
  --- Physical volume ---
  PV Name               /dev/sdc
  VG Name               vg_data
  PV Size               8.00 GiB / not usable 4.00 MiB
  Allocatable           yes
  PE Size               4.00 MiB
  Total PE              2047
  Free PE               1151
  Allocated PE          896
  PV UUID               d2Hkr0-Kntz-53M1-B7dY-Za3I-Y2ya-GisBlc
```

```
[root@server3 ~]# vgdisplay
  --- Volume group ---
  VG Name               vg_data
  System ID
  Format                lvm2
  Metadata Areas        1
  Metadata Sequence No  33
  VG Access             read/write
  VG Status             resizable
  MAX LV                0
  Cur LV                2
  Open LV               2
  Max PV                0
  Cur PV                1
  Act PV                1
  VG Size               <8.00 GiB
  PE Size               4.00 MiB
  Total PE              2047
  Alloc PE / Size       896 / 3.50 GiB
  Free  PE / Size       1151 / <4.50 GiB
  VG UUID               Ebj6CM-VCuo-zNeq-5yTx-ZZtK-ifhn-F2trDO
```

```
[root@server3 ~]# lvdisplay
  --- Logical volume ---
  LV Path                /dev/vg_data/lv_ingestion
  LV Name                lv_ingestion
  VG Name                vg_data
  LV UUID                QYJkPN-IMr3-xGbA-nNKX-e0ah-6ejg-LKWwHs
  LV Write Access        read/write
  LV Creation host, time server3.ex.com, 2025-05-07 02:15:25 +0200
  LV Status              available
  # open                 1
  LV Size                2.00 GiB
  Current LE             512
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     8192
  Block device           253:3

  --- Logical volume ---
  LV Path                /dev/vg_data/lv_analyses
  LV Name                lv_analyses
  VG Name                vg_data
  LV UUID                iI7lF3-gV3r-PPfd-C5o5-I7Il-H7NL-6vfYbn
  LV Write Access        read/write
  LV Creation host, time server3.ex.com, 2025-05-07 02:21:02 +0200
  LV Status              available
  # open                 1
  LV Size                1.50 GiB
  Current LE             384
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     8192
  Block device           253:4
```

(e) Screenshot of your cron table

```
[root@server3 ~]# crontab -l
29 11 * * * /root/sent/ingest.sh >> /var/log/ingest.log 2>&1
0 * * * * /root/sent/monitor_analyses.sh >> /var/log/monitor_analyses.log 2>&1
3 * * * * /root/sent/monitor_ingestion.sh >> /var/log/monitor_ingestion.log 2>&1
```

(f) Screenshot of all your scripts

ingest.sh

```
  GNU nano 2.9.8                                        /root/sent/ingest.sh

#!/bin/bash
#abdelrahman darwish 52-5930
# Set variables
DATE=$(date +%d-%m-%Y)
DEST_DIR=/root/ingestion/data
URL="https://github.com/Badr-AL101/rh2-project-csvs/blob/main/$DATE.csv"
CSV_FILE="$DEST_DIR/$DATE.csv"
COMPRESSED_FILE="$CSV_FILE.gz"

# Create directory if it doesn't exist
mkdir -p "$DEST_DIR"
# Check if mount exists
if ! mountpoint -q /root/ingestion; then
    echo "Ingestion directory is not mounted!"
    exit 1
fi

# Check disk usage running the script monitor_ingestion
monitor=$(bash /root/sent/monitor_ingestion.sh | tail -n 1 )
#if the script produces false cancel
if [ "$monitor" == "false" ]; then
    echo "insufficient space or an error ocuured. cancelling download"
    exit 1
fi
#the monitor script runned normally so print its result
echo "$monitor"
# Download the file using wget
wget -q -O "$CSV_FILE" "$URL"

# Verify download and compress
if [ -f "$CSV_FILE" ]; then
    gzip -f "$CSV_FILE"
    echo "File downloaded and compressed: $COMPRESSED_FILE"
else
    echo "Download failed. File not found: $CSV_FILE"
    exit 1
fi
```

monitor_ingestion

```
  GNU nano 2.9.8                                  /root/sent/monitor_ingestion.sh

#!/bin/bash
#abdelrahman darwish 52-5930
# This script monitors the disk usage of a specified mount point and automatically
# extends the logical volume and filesystem if the usage exceeds 90%.
# It is designed to work with ext4 filesystems using the `resize2fs` command.
# Define the logical volume path and mount directory
LV_PATH="/dev/vg_data/lv_ingestion"  # Replace with your actual logical volume path
MOUNT_DIR=/root/ingestion  # Replace with your actual mount directory
# Ensure the logical volume exists
if [ ! -e "$LV_PATH" ]; then
    echo "ERROR: Logical volume $LV_PATH does not exist."
    echo "false"
    exit 1
fi
# Ensure the mount directory exists
if [ ! -d "$MOUNT_DIR" ]; then
    echo "ERROR: Mount directory $MOUNT_DIR does not exist."
    echo "false"
    exit 1
fi
# Ensure the mount directory is actually mounted
if ! mountpoint -q "$MOUNT_DIR"; then
    echo "ERROR: $MOUNT_DIR is not a mount point."
    echo "false"
    exit 1
fi
# Get the current disk usage percentage of the mount point
USAGE=$(df --output=pcent "$MOUNT_DIR" | tail -1 | tr -dc '0-9')
# Check if the usage exceeds 90%
if [ "$USAGE" -ge 90 ]; then
    echo "Usage is $USAGE%. Attempting to extend LV..."
    # Get the current size of the logical volume in GB
    CURRENT_SIZE=$(lvs --noheadings --units g -o LV_SIZE "$LV_PATH" | tr -dc '0-9.')
    # Exit if the current size could not be determined
    if [ -z "$CURRENT_SIZE" ]; then
        echo "ERROR: Failed to get current LV size."
        echo "false"
```

```
    fi

    # Calculate 10% of the current size to determine the extension size
    EXTEND_BY=$(echo "$CURRENT_SIZE * 0.1" | bc)

    # Check if there is enough free space in the volume group to extend
    VG_NAME=$(lvs --noheadings -o VG_NAME "$LV_PATH" | tr -d ' ')
    FREE_SPACE=$(vgs --noheadings --units g -o VG_FREE "$VG_NAME" | tr -dc '0-9.')

    if (( $(echo "$EXTEND_BY > $FREE_SPACE" | bc -l) )); then
        echo "ERROR: Not enough free space in the volume group to extend."
        echo "false"
        exit 1
    fi

    # Extend the logical volume and grow the filesystem
    echo "Extending by ${EXTEND_BY}G..."
    lvextend -L +"${EXTEND_BY}G" "$LV_PATH"
    e2fsck -f "$LV_PATH" 2>/dev/null
    resize2fs "$LV_PATH"

    # Check if the extension and resizing were successful
    if [ $? -eq 0 ]; then
        echo "Extension successful."
        size=$(lvs --noheadings --units g -o LV_SIZE "$LV_PATH" | tr -dc '0-9.GMKT')
        echo "size was expanded to ${size} as it has reached 90 percent"
    else
        echo "ERROR: Failed to extend the logical volume or grow the filesystem."
        echo "false"
        exit 1
    fi
else
    # If usage is below 90%, no action is needed
    echo "Usage is $USAGE%. No extension needed."
    echo "size was not expanded as it has not reached 90 percent"
fi
```

monitor_analyses

```
  GNU nano 2.9.8                                  /root/sent/monitor_analyses.sh

#!/bin/bash
#abdelrahman darwish 52-5930
# This script monitors the disk usage of a specified mount point and automatically
# extends the logical volume and filesystem if the usage exceeds 90%.
# It is designed to work with ext4 filesystems using the `resize2fs` command.
# Define the logical volume path and mount directory
LV_PATH="/dev/vg_data/lv_analyses"
MOUNT_DIR=/root/analyses
# Check if the logical volume exists
if [ ! -e "$LV_PATH" ]; then
    echo "ERROR: Logical volume $LV_PATH does not exist."
    exit 1
fi
# Check if the mount directory exists
if [ ! -d "$MOUNT_DIR" ]; then
    echo "ERROR: Mount directory $MOUNT_DIR does not exist."
    exit 1
fi
# Check if the mount directory is actually mounted
if ! mountpoint -q "$MOUNT_DIR"; then
    echo "ERROR: $MOUNT_DIR is not mounted."
    exit 1
fi
# Get the current disk usage percentage of the mount point
USAGE=$(df --output=pcent "$MOUNT_DIR" | tail -1 | tr -dc '0-9')
# Check if the usage exceeds 90%
# If the usage is too high, attempt to extend the logical volume
if [ "$USAGE" -ge 90 ]; then
    echo "Usage is $USAGE%. Attempting to extend analyses LV..."
    # Get the current size of the logical volume in GB
    # This determines how much space is currently allocated
    CURRENT_SIZE=$(lvs --noheadings --units g -o LV_SIZE "$LV_PATH" | tr -dc '0-9.')
    # Exit if the current size could not be determined
    if [ -z "$CURRENT_SIZE" ]; then
        echo "ERROR: Could not determine current LV size."
        exit 1
    fi
```

```
    # Calculate 10% of the current size to determine the extension size
    # This calculates how much additional space to allocate
    EXTEND_BY=$(echo "$CURRENT_SIZE * 0.1" | bc)

    # Get the volume group name and free space available in the volume group
    # This ensures there is enough space to extend the logical volume
    VG_NAME=$(lvs --noheadings -o VG_NAME "$LV_PATH" | tr -d ' ')
    FREE_SPACE=$(vgs --noheadings --units g -o VG_FREE "$VG_NAME" | tr -dc '0-9.')

    # Check if there is enough free space in the volume group to extend
    if (( $(echo "$EXTEND_BY > $FREE_SPACE" | bc -l) )); then
        echo "ERROR: Not enough free space in the VG to extend."
        exit 1
    fi

    # Extend the logical volume and grow the filesystem
    # This increases the size of the logical volume and resizes the filesystem
    echo "Extending $LV_PATH by ${EXTEND_BY}G..."
    lvextend -L +"${EXTEND_BY}G" "$LV_PATH"
    e2fsck -f "$LV_PATH" 2>/dev/null
    resize2fs "$LV_PATH"

    # Check if the extension and resizing were successful
    if [ $? -eq 0 ]; then
        echo "Analyses LV successfully extended."
    else
        echo "ERROR: Extension or filesystem resize failed."
        exit 1
    fi
else
    # If usage is below 90%, no action is needed
    echo "Usage is $USAGE%. No extension needed."
fi
```

(g) semanage fcontext -l -C &(h) ls -Z /analyses/reports

```
[root@server3 ~]# semanage fcontext -l -C
SELinux fcontext                                    type               Context

/opt/VBoxGuestAdditions-7.1.6/other/mount.vboxsf    all files          system_u:object_r
:mount_exec_t:s0
/root/analyses/reports(/.*)?                        all files          system_u:object_r
:httpd_sys_content_t:s0
/usr/bin/VBoxClient                                 all files          system_u:object_r
:bin_t:s0
[root@server3 ~]# echo "testing final " > /root/analyses/reports/final
[root@server3 ~]# ls -Z /root/analyses/reports
unconfined_u:object_r:httpd_sys_content_t:s0 final
unconfined_u:object_r:httpd_sys_content_t:s0 test1.html
unconfined_u:object_r:httpd_sys_content_t:s0 test2
unconfined_u:object_r:httpd_sys_content_t:s0 test2.html
unconfined_u:object_r:httpd_sys_content_t:s0 test5.txt
unconfined_u:object_r:httpd_sys_content_t:s0 test.html
[root@server3 ~]#
```

(i) ls -al /ingestion/data

```
[root@server3 ~]# ls -al /root/ingestion/data
total 52
drwxr-xr-x. 2 root root  4096 May 13 16:46 .
drwxr-xr-x. 5 root root  4096 May 13 01:03 ..
-rwxr-xr-x. 1 root root    35 May  9 13:22 09-05-2025.csv.gz
-rwxr-xr-x. 1 root root    35 May 12 23:54 12-05-2025.csv.gz
-rwxr-xr-x. 1 root root 34807 May 13 16:46 13-05-2025.csv.gz
```